

Runtime Assurance for Autonomous Maritime Systems

A bounded safety governor for real-time supervision of autonomous vessel commands

Recommendation. Use A5, the evidence-conditioned predictive-filter hybrid, as the hackathon integration architecture. It gave the strongest operational result among the recovery-oriented candidates in the 60-episode development study: 99.83% gate acceptance and three deadline misses, compared with six for A3 and ten for A4. This is a prototype selection, not a certified safety conclusion; the formal study correctly withheld a winner because missions were censored, uncertainty bounds were exceeded, and only three seeds per scenario were run.

The problem and the safety boundary

An autonomous vessel can produce unsafe commands even when each onboard component appears healthy. Horizon places an independent runtime-assurance layer between the decision-making AI and the actuators. It continuously evaluates the proposed heading and speed against fused navigation state, obstacle estimates, uncertainty, platform limits, source health, and communication freshness. The AI retains control only while its command remains admissible. Otherwise, Horizon modifies the command, selects a validated recovery, or issues a minimum-risk fallback.

SENSE	UNDERSTAND	PROPOSE	ASSURE	ACT
Radar AIS GNSS IMU camera	Fusion tracks uncertainty health	External decision AI heading speed	A5 predictive + barrier checks	Exclusive gate watchdog vessel

Recommended architecture A5

A5 combines three ideas in one bounded decision path. First, it converts source health into explicit operating constraints: healthy evidence keeps nominal uncertainty and a 6 m/s cap; degraded evidence expands declared bounds by 1.5x and caps speed at 3 m/s; invalid evidence limits speed to 1 m/s and forces recovery. Second, it checks whether the proposed command remains safe through a short predictive handoff and a complete recovery continuation. Third, if prediction rejects the proposal, it invokes the A4 discrete plant-map barrier filter and validates any corrected command with the full 3DOF vessel rollout. Failure at any step closes safely to the recovery library or minimum-risk command.

- **Independent authority.** The decision AI never writes to actuators directly; only the exclusive gate can issue commands.
- **Fresh evidence.** The gate validates identity, epoch, lineage, deadline, source age, bounds, and final-command safety.
- **Perception health.** WaSR-T segmentation and neural representation health can enlarge margins or revoke camera-derived free-space authority; they never certify an empty sea alone.
- **Communications health.** Internal and inter-ship traffic contribute freshness, integrity, availability, consistency, and anomaly evidence without becoming actuator authority.

Safety contract Unknown, stale, unbounded, or late evidence cannot increase autonomy. It can only preserve or reduce authority.

Operational inputs and governed outputs

NAVIGATION AND PLATFORM	CONTACTS AND ENVIRONMENT	COMMUNICATIONS	AI AND PERCEPTION
GNSS, IMU, compass, speed log, actuator feedback, propulsion and steering limits	Radar, AIS, camera, depth, weather, sea state, charts and geofences	Internal bus and service traffic, inter-ship messages, age, loss, integrity and disagreement	Proposed command, plan, confidence, inference latency, WaSR-T outputs and representation health

Governed output. A lineage-bound pass, modified command, validated recovery, or minimum-risk fallback, followed by an independent gate receipt and watchdog status.

DEVELOPMENT EVIDENCE

What the 60-episode study established

The paired development matrix exercised A1-A5 across recoverable crossing, dense traffic, vessel degradation, and deliberately initially-unrecoverable encounters. Each architecture ran three seeds per scenario for 30 simulated seconds. The analyzer verified all 60 traces, diagnostic records, assumption audits, and the twelve complete five-candidate pairings.

60	0	15 / 15	0
complete episodes	recoverable collisions	unrecoverable collisions	unsafe or stale accepts

Candidate comparison

ID	Method	Gate accepted	Deadline misses	Decision for prototype
A1	Fixed CPA/TCPA threshold + Simplex	8.87%	0	Useful transparent baseline
A2	Gaussian collision-risk + Simplex	8.73%	1	Hold until calibrated
A3	Finite bounded predictive recovery	99.66%	6	Strong comparator
A4	Plant-map barrier + full rollout	99.44%	10	Strong filter; timing concern
A5	Evidence-conditioned predictive/barrier hybrid	99.83%	3	Recommended prototype

How to interpret the result

A5 is the best current implementation choice because it preserves explicit health-dependent limits, validates a recovery continuation, can fall back to a plant-aware barrier correction, and showed the best acceptance/miss balance among A3-A5. A1 and A2 were faster, but they mostly passed commands that the independent gate later rejected, so their apparent compute advantage did not translate into comparable closed-loop authority. A4 missed the deadline most often. A3 remained credible but had twice A5’s misses and slightly lower gate acceptance.

The evidence does not support a formal safety ranking. Every mission was censored at 30 seconds; 40 of 60 branches exceeded at least one declared engineering uncertainty bound; no independent sampled recovery boundary was included; modeled front-end latency was idealized; and gate/watchdog behavior materially influenced trajectories. All 15 collisions belonged to the deliberately unrecoverable stratum, so they do not measure preventable-collision performance.

Path to a defensible selection

- **Freeze the protocol.** Calibrate A2 and perception-health thresholds without reading held-out outcomes; lock code, models, bounds, and timing profiles.
- **Run the required study.** Use at least 1,000 held-out episodes with 30 or more paired seeds per stochastic scenario and architecture cell, plus explicit recoverability strata.
- **Measure the full loop.** Include sensor, network, AI, fusion, assurance, gate, and actuator latency on target hardware; characterize deadline misses and watchdog takeover.
- **Qualify each operating mode.** Validate marine physics, uncertainty coverage, degraded sensing, communication faults, and real vessel endpoints before expanding authority.

Architecture decision Proceed with A5 for the hackathon demo and preserve A1 as the transparent baseline. Keep the exclusive gate and watchdog independent of either candidate.

Evidence source: Horizon A32 A1-A5 30-second development analysis, source revision 28eb923, analysis hash 8 004 358ed9ac50c824 750a6906e344b94e438ec04b0448bca421e7 768 072e909. Development evidence only.